

Forces That Keep Things Turning: Circular Motion

SimuPhysics Circular Motion Simulation · SimuPhysics

Years 7 and 10

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

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Simulation: <https://simuphysics.com/view/mechanics-circular-motion-simulation.html>

Curriculum links: AC9S7U04 · AC9S10U05 · forces · motion

Learning intentions

- I can describe the direction of the force that keeps an object moving in a circle.
- I can predict how changing radius or speed affects centripetal force.
- I can use the relationship between force, mass and acceleration to explain circular motion quantitatively.

Getting started

Open the SimuPhysics Circular Motion Simulation. Start with a default radius and speed and let the object move around its circular path, watching the direction of the force arrow shown on the object. Use the sliders or input fields to change the radius and speed for the tasks below, and check your predictions against the live readout each time.

Tasks

1. Watch the object move around the circular path. In which direction does the force arrow point relative to the object's motion — towards the centre, away from the centre, or in the direction of travel?

2. Explain in your own words why this force is needed to keep the object moving in a circle rather than in a straight line.

3. Increase the speed of the object while keeping the radius the same. Predict what will happen to the centripetal force before you check, then record what the readout shows.

4. Now keep the speed the same as your original setting but increase the radius. Predict what will happen to the centripetal force, then record the readout.

5. Keep the mass constant and test four different combinations of radius and speed. Record the force reading each time.

Radius (m)	Speed (m/s)	Centripetal force reading

6. Using your table, describe the relationship between speed and centripetal force, and separately, the relationship between radius and centripetal force.

7. Choose one trial from your table. Using $F = mv^2/r$ (with an assumed reasonable mass value, e.g. 1 kg, if the simulation doesn't state one), calculate the centripetal force by hand and compare it with the simulation's reading.

Working:

8. A car takes a corner too fast on a wet road and skids outward. Using what you've learned about centripetal force in this simulation, explain what happened in terms of the force needed to keep the car turning.

Extension (optional)

Roller coasters use circular loops that are often taller and narrower at the top than a perfect circle. Use your understanding of the relationship between speed, radius and centripetal force to suggest why this shape might help keep riders safely in their seats.
